

## PUNTZI MOUNTAIN, BC, Canada

WMO: 710500

Lat: 52.1144N

Lon: 124.1356W

Elev: 2985

StdP: 13.18

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |          | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|-------------|-------------|-----------------------|----------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%          |             |                       |          |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD     |              |
| (a)<br>12     | (b)<br>-28.1 | (c)<br>-18.4 | (d)<br>-33.8                  | (e)<br>0.9 | (f)<br>-26.1 | (g)<br>-25.0 | (h)<br>1.5 | (i)<br>-17.1 | (j)<br>16.4           | (k)<br>34.5 | (l)<br>13.8 | (m)<br>31.8 | (n)<br>0.5            | (o)<br>0 | (p)<br>0.648 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 30.2                   | 85.0            | 58.3 | 81.2 | 56.9 | 77.4 | 55.5 | 59.9                | 79.9 | 58.3 | 76.9 | 56.8 | 73.9 | 6.2                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 53.4                            | 67.7 | 61.0 | 51.2 | 62.5 | 60.0 | 49.2 | 57.8 | 59.4 | 27.6          | 80.1 | 26.5 | 76.8 | 25.5 | 74.2 | 66.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 15.7              | 13.3 | 11.2 | -37.5                      | 90.5 | 8.9                | 3.0 | -43.9                                              | 92.7 | -49.2      | 94.4 | -54.2      | 96.1 | -60.7      | 98.3 |
|                   |      |      | -36.0                      | 62.4 | 7.0                | 2.0 | -41.1                                              | 63.9 | -45.2      | 65.0 | -49.1      | 66.1 | -54.2      | 67.6 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 36.6   | 12.3  | 17.6  | 28.1  | 38.1  | 47.6  | 53.6  | 59.0  | 57.9  | 49.9  | 38.2  | 24.1  | 11.8  |
|                                                                                  | DBStd       | 18.84  | 13.39 | 11.46 | 9.91  | 6.06  | 6.45  | 5.38  | 5.47  | 5.62  | 6.16  | 6.75  | 10.57 | 11.91 |
|                                                                                  | HDD50       | 5676   | 1168  | 908   | 679   | 360   | 125   | 23    | 3     | 6     | 76    | 368   | 777   | 1183  |
|                                                                                  | HDD65       | 10383  | 1633  | 1328  | 1144  | 808   | 541   | 343   | 198   | 229   | 454   | 830   | 1227  | 1648  |
|                                                                                  | CDD50       | 789    | 0     | 0     | 0     | 2     | 49    | 131   | 282   | 250   | 72    | 3     | 0     | 0     |
|                                                                                  | CDD65       | 21     | 0     | 0     | 0     | 0     | 0     | 1     | 12    | 8     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH74       | 1592   | 0     | 0     | 0     | 3     | 69    | 174   | 668   | 561   | 114   | 3     | 0     | 0     |
|                                                                                  | CDH80       | 433    | 0     | 0     | 0     | 0     | 9     | 41    | 212   | 153   | 18    | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.3    | 2.0   | 2.5   | 3.7   | 4.3   | 4.4   | 4.5   | 4.0   | 3.4   | 3.1   | 3.0   | 2.5   | 2.1   |
| Precipitation                                                                    | PrecAvg     | 36.1   | 4.1   | 2.6   | 3.0   | 2.2   | 2.0   | 2.7   | 2.0   | 2.0   | 2.4   | 3.8   | 5.0   | 4.3   |
|                                                                                  | PrecMax     | 46.3   | 8.5   | 5.8   | 5.2   | 4.0   | 3.9   | 5.4   | 4.7   | 5.4   | 5.0   | 7.0   | 10.4  | 6.3   |
|                                                                                  | PrecMin     | 28.3   | 2.2   | 0.1   | 0.6   | 0.9   | 0.8   | 1.3   | 0.4   | 0.5   | 0.3   | 0.4   | 2.0   | 2.0   |
|                                                                                  | PrecStd     | 4.6    | 1.4   | 1.3   | 1.1   | 0.8   | 0.8   | 0.9   | 1.0   | 1.1   | 1.3   | 1.6   | 2.1   | 1.1   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 43.5  | 46.9  | 59.1  | 70.7  | 80.4  | 85.6  | 90.9  | 88.6  | 82.8  | 69.9  | 52.1  | 41.4  |
|                                                                                  |             | MCWB   | 36.6  | 37.1  | 44.8  | 49.8  | 53.5  | 57.5  | 60.7  | 59.8  | 56.6  | 52.0  | 43.0  | 35.3  |
|                                                                                  | 2%          | DB     | 38.1  | 42.6  | 51.8  | 62.5  | 75.0  | 79.0  | 85.7  | 84.3  | 77.3  | 62.0  | 44.9  | 36.5  |
|                                                                                  |             | MCWB   | 34.0  | 35.0  | 39.8  | 46.2  | 51.9  | 54.7  | 58.8  | 58.6  | 55.4  | 48.4  | 38.9  | 32.8  |
|                                                                                  | 5%          | DB     | 34.0  | 39.3  | 47.5  | 57.3  | 69.8  | 74.1  | 81.5  | 80.5  | 72.2  | 56.3  | 40.5  | 32.4  |
|                                                                                  |             | MCWB   | 31.3  | 32.8  | 37.6  | 43.3  | 50.4  | 53.6  | 57.4  | 57.4  | 54.0  | 45.2  | 35.8  | 29.8  |
|                                                                                  | 10%         | DB     | 30.3  | 35.3  | 43.8  | 53.2  | 65.2  | 69.6  | 77.1  | 76.3  | 66.9  | 51.8  | 36.7  | 28.6  |
|                                                                                  |             | MCWB   | 28.3  | 30.3  | 35.3  | 40.6  | 48.4  | 51.8  | 56.1  | 55.9  | 51.6  | 42.6  | 33.7  | 26.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 37.5  | 38.2  | 45.4  | 50.3  | 55.2  | 58.7  | 62.4  | 62.3  | 58.7  | 52.8  | 43.5  | 36.2  |
|                                                                                  |             | MCDB   | 42.0  | 45.0  | 57.6  | 68.5  | 74.5  | 77.4  | 85.0  | 82.0  | 76.5  | 68.1  | 50.4  | 40.5  |
|                                                                                  | 2%          | WB     | 34.4  | 35.4  | 40.7  | 47.1  | 53.0  | 56.5  | 60.1  | 60.1  | 56.6  | 49.5  | 39.6  | 33.1  |
|                                                                                  |             | MCDB   | 37.4  | 41.6  | 50.2  | 60.5  | 71.0  | 74.4  | 80.7  | 79.9  | 73.9  | 59.8  | 44.0  | 36.1  |
|                                                                                  | 5%          | WB     | 31.7  | 33.2  | 38.2  | 44.0  | 51.3  | 54.7  | 58.4  | 58.4  | 54.6  | 46.2  | 36.5  | 30.1  |
|                                                                                  |             | MCDB   | 33.8  | 38.4  | 46.5  | 56.0  | 67.3  | 70.7  | 77.3  | 77.1  | 70.1  | 55.2  | 39.6  | 32.5  |
|                                                                                  | 10%         | WB     | 28.4  | 30.9  | 35.8  | 41.3  | 49.4  | 52.8  | 56.9  | 56.8  | 52.4  | 43.4  | 33.8  | 26.7  |
|                                                                                  |             | MCDB   | 30.5  | 35.1  | 43.1  | 51.5  | 63.1  | 67.2  | 73.9  | 73.5  | 65.5  | 50.6  | 36.3  | 28.6  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 23.4  | 30.3  | 27.5  | 27.5  | 29.1  | 27.5  | 30.2  | 31.8  | 30.5  | 25.6  | 19.4  | 21.5  |
|                                                                                  |             | MCDBR  | 25.3  | 32.1  | 29.6  | 35.7  | 39.1  | 37.2  | 40.3  | 40.5  | 42.0  | 33.9  | 23.1  | 24.0  |
|                                                                                  |             | MCWBR  | 21.5  | 24.8  | 18.8  | 20.4  | 19.2  | 17.0  | 17.3  | 18.5  | 22.1  | 21.0  | 16.7  | 20.6  |
|                                                                                  | 5% WB       | MCDBR  | 23.6  | 29.9  | 27.8  | 33.0  | 34.2  | 31.7  | 35.1  | 36.2  | 37.6  | 31.7  | 20.8  | 23.4  |
|                                                                                  |             | MCWBR  | 20.1  | 23.2  | 17.9  | 19.3  | 17.3  | 15.1  | 15.9  | 17.1  | 20.2  | 20.2  | 15.4  | 20.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.275 | 0.281 | 0.302 | 0.327 | 0.354 | 0.351 | 0.341 | 0.342 | 0.312 | 0.297 | 0.286 | 0.269 |
|                                                                                  | taud        |        | 2.350 | 2.437 | 2.446 | 2.414 | 2.374 | 2.417 | 2.492 | 2.483 | 2.566 | 2.569 | 2.427 | 2.325 |
|                                                                                  | Ebn at Noon |        | 234   | 271   | 285   | 289   | 285   | 286   | 287   | 281   | 280   | 263   | 229   | 214   |
|                                                                                  | Edh at Noon |        | 19    | 23    | 28    | 33    | 36    | 35    | 32    | 30    | 25    | 20    | 17    | 16    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 284   | 577   | 967   | 1426  | 1697  | 1786  | 1880  | 1604  | 1113  | 647   | 315   | 222   |
|                                                                                  | RadStd      |        | 26    | 57    | 75    | 110   | 125   | 151   | 179   | 81    | 103   | 67    | 34    | 21    |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (4 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +121  | N/A   |

Nomenclature: See separate page